

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Developed and maintained **automated test scripts** using **Python** for embedded software components, increasing test coverage by **30%** and reducing manual regression testing effort by **40%**.
- Executed comprehensive test plans for **backend applications** and **networking protocols**, identifying and documenting over **150 defects** with detailed root cause analysis, ensuring adherence to aerospace-grade reliability standards.
- Collaborated within an **Agile environment** to analyze requirements and technical designs, contributing to test scope definition and strategy for mission-critical **Free Space Optics (FSO)** systems.
- Integrated automated test suites into **CI/CD pipelines** using **Jenkins**, enabling continuous software quality assurance for embedded systems deployed in satellite communications.
- Performed production support and incident response for enterprise applications, managing SLAs and resolving P1/P2 incidents for critical financial-services operations dashboards, maintaining **99.9% uptime**.
- Supported and integrated with enterprise business systems, including **ERP modules**, managing change control processes and coordinating User Acceptance Testing (UAT) for system configuration updates across interconnected platforms.

SKILLS

Programming Languages – Python, Java, SQL, C, C++

Testing & QA – Test Planning, Test Execution, Automated Testing, CI/CD, Jenkins, BitBucket Pipelines, Agile Methodologies, Defect Tracking, Root Cause Analysis

Embedded & Backend – Embedded Software, Backend Applications, REST APIs, Spring Boot, Linux

Networking – Networking Protocols, Free Space Optics (FSO)

Cloud & DevOps – AWS, Docker, Git

Data Analytics & BI – Power BI, Tableau, DAX, Power Query, Data Modelling, SQL

Enterprise Systems – ERP Support, Change Management

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Space-Grade Embedded Systems Test Automation Framework

- Designed and implemented a comprehensive test automation framework for **embedded software** and **backend applications** targeting **Free Space Optics (FSO)** terminals, achieving **85% automated test coverage**.

- Developed **Python** test scripts to simulate complex **networking protocols** and hardware-in-the-loop scenarios, reducing test cycle time by **50%** and ensuring system resilience for satellite communications.
- Integrated automated test execution into **CI/CD pipelines** using **Jenkins** and **BitBucket Pipelines**, providing rapid feedback on software quality and facilitating continuous integration for mission-critical deployments.

Financial Services Transaction Monitoring Dashboard

- Enhanced a payments operations dashboard by implementing real-time **transaction monitoring** and alerting for failed payment messages, improving issue detection by **25%**.
- Developed automated Python scripts for data validation and reconciliation workflows within the financial-services domain, ensuring data integrity across multiple database systems (PostgreSQL, Oracle).
- Supported production environments with strict SLAs and audit trail requirements, contributing to **GDPR-aware** operational processes and cross-functional collaboration with business analysts.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)